

SVM Employment Analysis

Predicting 2023 Employment from 2021 Predictors

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```
DATA_DIR <- "J362299"
DATA <- file.path(DATA_DIR, "tas_analysis.csv")

PREDICTORS <- c("female", "age_2021", "emp_2021_working", "educ_yrs_2021",
               "days_unemp_2021", "flourish_2021", "soc_anxiety_2021",
               "log_fam_expend_2021")

OUTCOME <- "employed"
CLUSTER <- "id_fam1968"
```

```
load_data <- function(path = DATA) {
  df <- read.csv(path)
  for (col in PREDICTORS) {
    if (any(is.na(df[[col]]))) {
      df[[paste0(col, "_miss")]] <- as.integer(is.na(df[[col]]))
      df[[col]][is.na(df[[col]])] <- median(df[[col]], na.rm = TRUE)
    }
  }
  df
}
```

```
auc_roc <- function(y, p) {
  r <- rank(p); n1 <- sum(y == 1); n0 <- sum(y == 0)
  (sum(r[y == 1]) - n1 * (n1 + 1) / 2) / (n1 * n0)
}
```

```
auc_pr <- function(y, p) {
  o <- order(p, decreasing = TRUE); y <- y[o]
  tp <- cumsum(y == 1); fp <- cumsum(y == 0); P <- sum(y == 1)
  precision <- tp / (tp + fp); recall <- tp / P
  sum(diff(c(0, recall)) * precision)
}
```

```
roc_curve_data <- function(y, p) {
  o <- order(p, decreasing = TRUE); y <- y[o]
  n1 <- sum(y == 1); n0 <- sum(y == 0)
  data.frame(fpr = c(0, cumsum(y == 0) / n0),
             tpr = c(0, cumsum(y == 1) / n1))
}
```

```

}

pr_curve_data <- function(y, p) {
  o <- order(p, decreasing = TRUE); y <- y[o]
  P <- sum(y == 1)
  tp <- cumsum(y == 1); fp <- cumsum(y == 0)
  data.frame(recall = c(0, tp / P),
             precision = c(1, tp / (tp + fp)))
}

df <- load_data()
feat_cols <- c(PREDICTORS, grep("_miss$", names(df), value = TRUE))

```

Data Overview

```
cat(sprintf("- **Observations:** %d\n", nrow(df)))
```

- Observations: 1478

```
cat(sprintf("- **Outcome base rate (employed = 1):** %.3f\n", mean(df[[OUTCOME]])))
```

- Outcome base rate (employed = 1): 0.806

```
cat(sprintf("- **Family clusters:** %d\n", length(unique(df[[CLUSTER]]))))
```

- Family clusters: 831

```
cat(sprintf("- **Features used:** %s\n", paste(feat_cols, collapse = ", ")))
```

- Features used: female, age_2021, emp_2021_working, educ_yrs_2021, days_unemp_2021, flourish_2021, soc_anxiety_2021, log_fam_expend_2021, educ_yrs_2021_miss, days_unemp_2021_miss, flourish_2021_miss, soc_anxiety_2021_miss

1. SVM — RBF Kernel

Family-grouped 5-fold CV; class weights inversely proportional to class frequency.

```
X      <- as.matrix(df[, feat_cols])
y      <- df[[OUTCOME]]
groups <- as.character(df[[CLUSTER]])

X_scaled <- scale(X)

set.seed(0)
fams     <- unique(groups)
fam_fold <- setNames(sample(rep(1:5, length.out = length(fams))), fams)
fold_id  <- fam_fold[groups]

pos_w    <- sum(y == 0) / sum(y == 1)
class_weights <- c("0" = 1, "1" = pos_w)

oof_prob_rbf <- numeric(length(y))
oof_class_rbf <- integer(length(y))

for (k in 1:5) {
  tr <- fold_id != k
  te <- fold_id == k
  m  <- svm(
    x      = X_scaled[tr, , drop = FALSE],
    y      = as.factor(y[tr]),
    kernel = "radial",
    probability = TRUE,
    class.weights = class_weights,
    cost       = 1,
    gamma      = 1 / ncol(X_scaled)
  )
  preds      <- predict(m, X_scaled[te, , drop = FALSE], probability = TRUE)
  oof_prob_rbf[te] <- attr(preds, "probabilities")[, "1"]
  oof_class_rbf[te] <- as.integer(as.character(preds))
}

roc_auc_rbf <- auc_roc(y, oof_prob_rbf)
pr_auc_rbf  <- auc_pr(y, oof_prob_rbf)
cm_rbf      <- table(Predicted = oof_class_rbf, Actual = y)
```

Metrics

```
cat(sprintf("- **ROC-AUC:** %.3f\n", roc_auc_rbf))
```

- ROC-AUC: 0.744

```
cat(sprintf("- **PR-AUC:** %.3f (base rate: %.3f)\n", pr_auc_rbf, mean(y)))
```

- PR-AUC: 0.908 (base rate: 0.806)

Confusion Matrix (OOF):

```
knitr::kable(cm_rbf)
```

	0	1
0	21	13
1	265	1179

Permutation Feature Importance

```
m_full_rbf <- svm(
  x = X_scaled, y = as.factor(y),
  kernel = "radial", probability = TRUE,
  class.weights = class_weights, cost = 1, gamma = 1 / ncol(X_scaled)
)
base_auc_rbf <- auc_roc(
  y, attr(predict(m_full_rbf, X_scaled, probability = TRUE), "probabilities")[, "1"]
)
imp_rbf <- sapply(seq_along(feat_cols), function(j) {
  Xp <- X_scaled; Xp[, j] <- sample(Xp[, j])
  base_auc_rbf - auc_roc(
    y, attr(predict(m_full_rbf, Xp, probability = TRUE), "probabilities")[, "1"]
  )
})
imp_df_rbf <- data.frame(feature = feat_cols, importance = imp_rbf)
imp_df_rbf <- imp_df_rbf[order(imp_df_rbf$importance, decreasing = TRUE), ]

knitr::kable(imp_df_rbf, digits = 4,
  col.names = c("Feature", "Drop in ROC-AUC"))
```

	Feature	Drop in ROC-AUC
3	emp_2021_working	0.1334
8	log_fam_expend_2021	0.0543
4	educ_yrs_2021	0.0540
7	soc_anxiety_2021	0.0162
11	flourish_2021_miss	0.0150
10	days_unemp_2021_miss	0.0145
1	female	0.0134
2	age_2021	0.0131
9	educ_yrs_2021_miss	0.0094
12	soc_anxiety_2021_miss	0.0084
6	flourish_2021	0.0073
5	days_unemp_2021	0.0033

Plots

```
rd_rbf <- roc_curve_data(y, oof_prob_rbf)
p_roc_rbf <- ggplot(rd_rbf, aes(fpr, tpr)) +
  geom_line(color = "#2166ac", linewidth = 1) +
  geom_abline(linetype = "dashed", color = "grey60") +
  annotate("text", x = 0.6, y = 0.15,
    label = sprintf("AUC = %.3f", roc_auc_rbf), size = 4.5) +
  labs(title = "ROC Curve", x = "False Positive Rate", y = "True Positive Rate") +
  theme_minimal(base_size = 12)
```

```

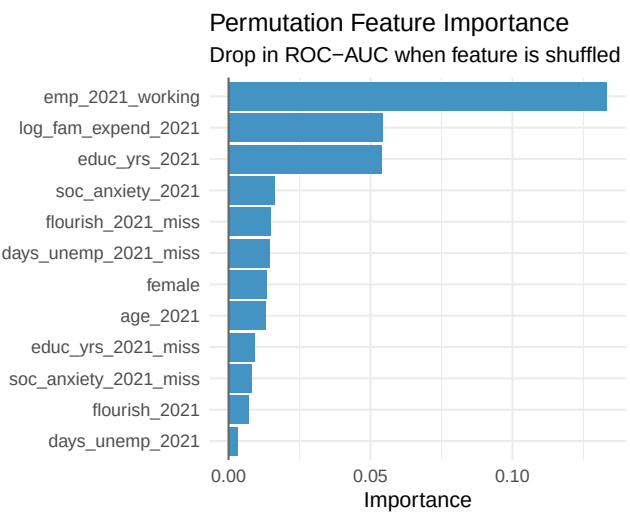
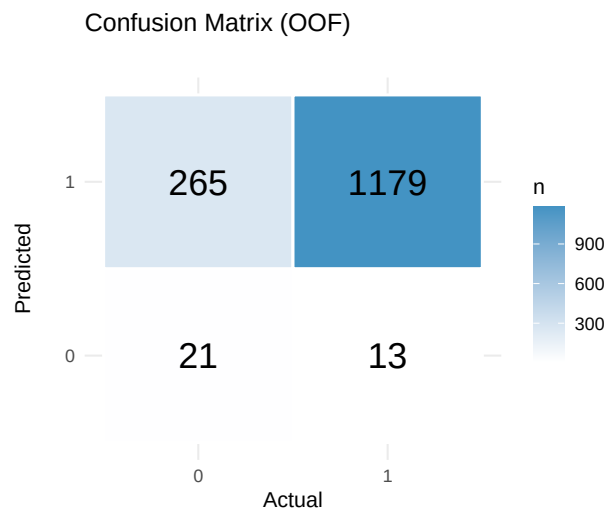
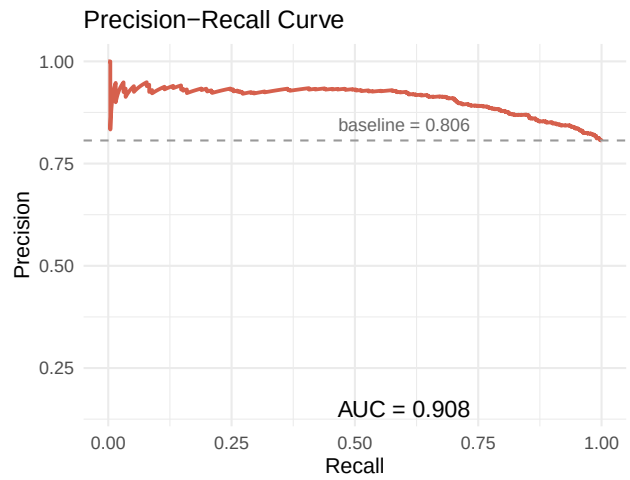
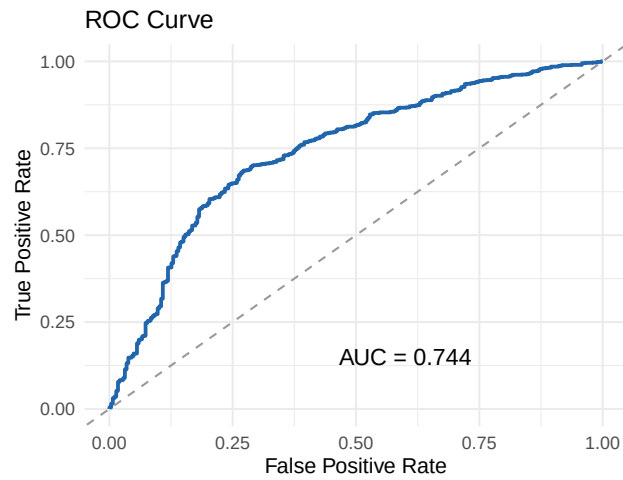
prd_rbf <- pr_curve_data(y, oof_prob_rbf)
p_pr_rbf <- ggplot(prd_rbf, aes(recall, precision)) +
  geom_line(color = "#d6604d", linewidth = 1) +
  geom_hline(yintercept = mean(y), linetype = "dashed", color = "grey60") +
  annotate("text", x = 0.6, y = mean(y) + 0.04,
    label = sprintf("baseline = %.3f", mean(y)), size = 3.5, color = "grey40") +
  annotate("text", x = 0.6, y = 0.15,
    label = sprintf("AUC = %.3f", pr_auc_rbf), size = 4.5) +
  labs(title = "Precision-Recall Curve", x = "Recall", y = "Precision") +
  theme_minimal(base_size = 12)

cm_df_rbf <- as.data.frame(cm_rbf)
p_cm_rbf <- ggplot(cm_df_rbf, aes(x = factor(Actual), y = factor(Predicted), fill = Freq)) +
  geom_tile(color = "white", linewidth = 0.8) +
  geom_text(aes(label = Freq), size = 7) +
  scale_fill_gradient(low = "white", high = "#4393c3") +
  labs(title = "Confusion Matrix (OOF)", x = "Actual", y = "Predicted", fill = "n") +
  theme_minimal(base_size = 12)

p_imp_rbf <- ggplot(imp_df_rbf, aes(x = reorder(feature, importance), y = importance)) +
  geom_col(fill = "#4393c3") +
  geom_hline(yintercept = 0, color = "grey40") +
  coord_flip() +
  labs(title = "Permutation Feature Importance",
    subtitle = "Drop in ROC-AUC when feature is shuffled",
    x = NULL, y = "Importance") +
  theme_minimal(base_size = 12)

(p_roc_rbf | p_pr_rbf) / (p_cm_rbf | p_imp_rbf)

```



2. Softmax Classifier (Ridge Logistic Regression)

Ridge-penalised logistic regression; tuned by nested 5-fold CV inside each outer fold.

```
obs_w <- ifelse(y == 1, sum(y == 0) / sum(y == 1), 1.0)

oof_prob_sm <- numeric(length(y))
oof_class_sm <- integer(length(y))

for (k in 1:5) {
  tr <- fold_id != k
  te <- fold_id == k

  cv_fit <- cv.glmnet(X_scaled[tr, , drop = FALSE], y[tr],
                     family = "binomial", alpha = 0,
                     weights = obs_w[tr], nfolds = 5)
  probs <- predict(cv_fit, X_scaled[te, , drop = FALSE],
                  s = cv_fit$lambda.min, type = "response")
  oof_prob_sm[te] <- as.vector(probs)
  oof_class_sm[te] <- as.integer(oof_prob_sm[te] > 0.5)
}

roc_auc_sm <- auc_roc(y, oof_prob_sm)
pr_auc_sm <- auc_pr(y, oof_prob_sm)
cm_sm <- table(Predicted = oof_class_sm, Actual = y)

cv_full <- cv.glmnet(X_scaled, y, family = "binomial", alpha = 0, weights = obs_w)
m_full_sm <- glmnet(X_scaled, y, family = "binomial", alpha = 0,
                  lambda = cv_full$lambda.min, weights = obs_w)
coef_vals <- as.vector(coef(m_full_sm))[-1]
coef_df <- data.frame(feature = feat_cols, coefficient = coef_vals)
coef_df <- coef_df[order(abs(coef_df$coefficient), decreasing = TRUE), ]
```

Metrics

```
cat(sprintf("- **ROC-AUC:** %.3f\n", roc_auc_sm))
```

- ROC-AUC: 0.760

```
cat(sprintf("- **PR-AUC:** %.3f (base rate: %.3f)\n", pr_auc_sm, mean(y)))
```

- PR-AUC: 0.914 (base rate: 0.806)

Confusion Matrix (OOF):

```
knitr::kable(cm_sm)
```

	0	1
0	200	332
1	86	860

Coefficients (Full-Data Fit, Ridge-Penalised)

```
knitr::kable(coef_df, digits = 4,
             col.names = c("Feature", "Coefficient"))
```

	Feature	Coefficient
3	emp_2021_working	0.5454
4	educ_yrs_2021	0.4106
8	log_fam_expend_2021	0.4092
7	soc_anxiety_2021	0.2002
9	educ_yrs_2021_miss	0.1151
6	flourish_2021	0.0647
5	days_unemp_2021	-0.0570
11	flourish_2021_miss	0.0557
1	female	-0.0458
2	age_2021	0.0336
10	days_unemp_2021_miss	-0.0233
12	soc_anxiety_2021_miss	0.0005

Plots

```
rd_sm <- roc_curve_data(y, oof_prob_sm)
p_roc_sm <- ggplot(rd_sm, aes(fpr, tpr)) +
  geom_line(color = "#2166ac", linewidth = 1) +
  geom_abline(linetype = "dashed", color = "grey60") +
  annotate("text", x = 0.6, y = 0.15,
    label = sprintf("AUC = %.3f", roc_auc_sm), size = 4.5) +
  labs(title = "ROC - Softmax (Ridge Logistic)",
    x = "False Positive Rate", y = "True Positive Rate") +
  theme_minimal(base_size = 12)

prd_sm <- pr_curve_data(y, oof_prob_sm)
p_pr_sm <- ggplot(prd_sm, aes(recall, precision)) +
  geom_line(color = "#d6604d", linewidth = 1) +
  geom_hline(yintercept = mean(y), linetype = "dashed", color = "grey60") +
  annotate("text", x = 0.6, y = 0.15,
    label = sprintf("AUC = %.3f", pr_auc_sm), size = 4.5) +
  labs(title = "PR - Softmax (Ridge Logistic)",
    x = "Recall", y = "Precision") +
  theme_minimal(base_size = 12)

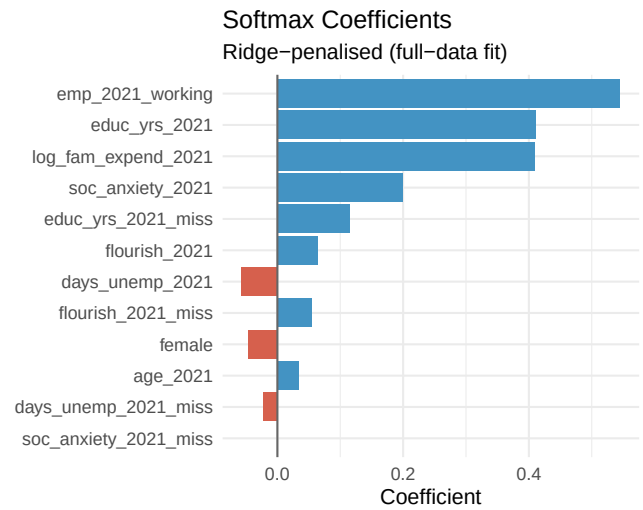
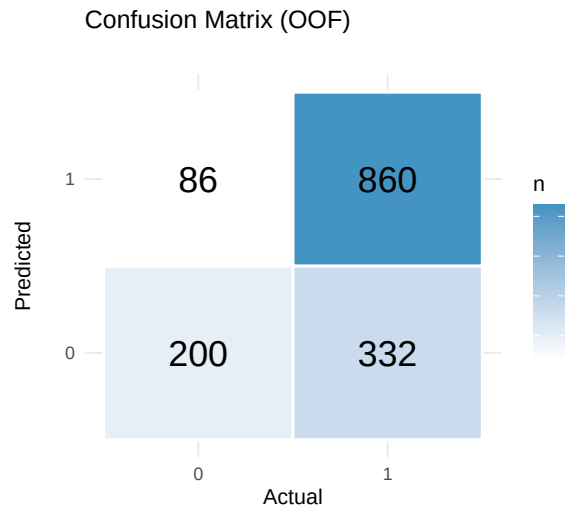
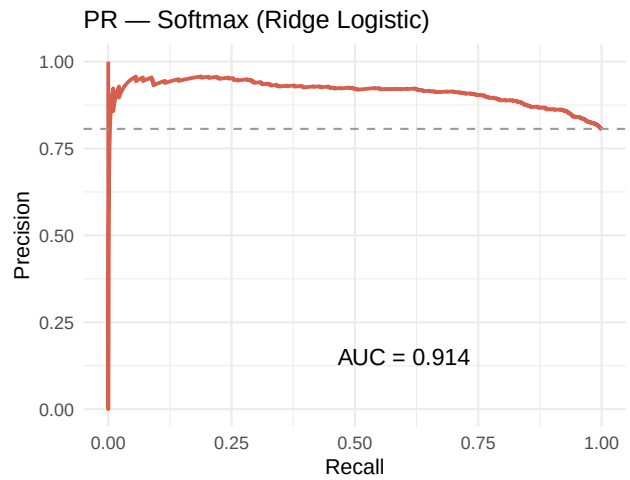
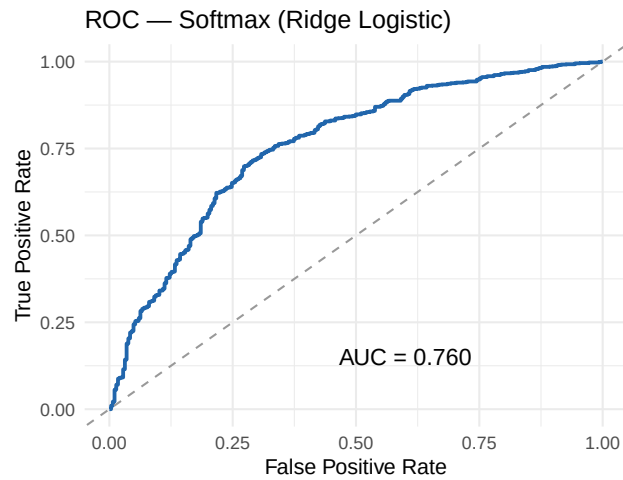
cm_df_sm <- as.data.frame(cm_sm)
p_cm_sm <- ggplot(cm_df_sm, aes(x = factor(Actual), y = factor(Predicted), fill = Freq)) +
  geom_tile(color = "white", linewidth = 0.8) +
  geom_text(aes(label = Freq), size = 7) +
  scale_fill_gradient(low = "white", high = "#4393c3") +
  labs(title = "Confusion Matrix (OOF)", x = "Actual", y = "Predicted", fill = "n") +
  theme_minimal(base_size = 12)

p_coef_sm <- ggplot(coef_df, aes(x = reorder(feature, abs(coefficient)), y = coefficient)) +
  geom_col(aes(fill = coefficient > 0)) +
  scale_fill_manual(values = c("TRUE" = "#4393c3", "FALSE" = "#d6604d"), guide = "none") +
  geom_hline(yintercept = 0, color = "grey40") +
  coord_flip() +
  labs(title = "Softmax Coefficients",
    subtitle = "Ridge-penalised (full-data fit)",
    x = NULL, y = "Coefficient") +
```



```
theme_minimal(base_size = 12)
```

```
(p_roc_sm | p_pr_sm) / (p_cm_sm | p_coef_sm)
```



3. Model Comparison

```
cat(sprintf("| Model | ROC-AUC | PR-AUC |\n"))
```

```
Model | ROC-AUC | PR-AUC |
```

```
cat(sprintf("|---|---|---|\n"))
```

```
|---|---|
```

```
cat(sprintf("| SVM RBF kernel | %.3f | %.3f |\n", roc_auc_rbf, pr_auc_rbf))
```

```
SVM RBF kernel | 0.744 | 0.908 |
```

```
cat(sprintf("| Softmax (ridge) | %.3f | %.3f |\n", roc_auc_sm, pr_auc_sm))
```

```
Softmax (ridge) | 0.760 | 0.914 |
```